MEGADOOR

ASSA ABLOY

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Calculation of uniformly loaded intermediate beam

Project: **ARRF5 Delhi Fire Station**
Project number/Door number: 22240 - 22241
Date: 210930
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Profile type: 160x70 -100

Nominal differential pressure: $P := 1150 \cdot \frac{N}{m^2}$

Fabric contribution: $cf := 1.041$

Reduction factor due to profile rotation: $rf := 0.975$

Actual pressure to be supported by intermediate beam: $p := \frac{P}{cf}$ $p = 1104.707 \ \frac{N}{m^2}$

Moment of inertia: $I := rf \cdot 5280000 \cdot mm^4$ $I = 5148000 \ mm^4$

Support distance: $l := 8760 \cdot mm$

Plastic section modulus: $Z := rf \cdot 78100 \cdot mm^3$ $Z = 76147.5 \ mm^3$

Yield point: $\sigma_y := 240 \cdot \frac{N}{mm^2}$

Modulus of elasticity: $E := 70000 \cdot \frac{N}{mm^2}$

Safety factor acc. to EN 12604: $n_s := 1.5$

Allowable stress: $\sigma_a := \frac{\sigma_y}{n_s}$ $\sigma_a = 160 \ \frac{N}{mm^2}$

Max allowed beam spacing: $d. := \frac{\sigma_a \cdot 8 \cdot Z}{l^2 \cdot p}$ $d. = 1149.8 \ mm$

Actual spacing: $d := 898 \cdot mm$

Beam stress at actual spacing: $\sigma := \frac{d}{\left(\frac{8 \cdot Z}{l^2 \cdot p}\right)}$ $\sigma = 125 \ \frac{N}{mm^2}$

Apparent shortening:

$$\Delta := \int_{0 \cdot mm}^{l} \sqrt{1 + \left(\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{5 \cdot d \cdot l \cdot p \cdot l^3}{384 \cdot E \cdot I} \cdot \left(1 - \frac{4 \cdot x^2}{l^2}\right) \cdot \left(1 - \frac{4 \cdot x^2}{5 \cdot l^2}\right)\right)\right)^2} \, \mathrm{d}x - l$$

$\Delta = 12.634 \ mm$

Deflection: $\delta := \frac{5 \cdot d \cdot p \cdot l^4}{384 \cdot E \cdot I}$ $\delta = 211 \ mm$

Deflection limits are L/30 for aluminum and L/120 for steel.

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